

CE3420 - GIS Assignment 2 Report

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Overview

This assignment covers Python fundamentals for GIS applications, including distance calculations using various methods, working with web APIs, and data manipulation with CSV files and Pandas.

Tasks Completed

1. The Python Standard Library

Objective: Calculate the distance between two cities using the Haversine formula.

- **Cities:** Hyderabad (17.4065, 78.4772) and Bengaluru (12.9629, 77.5775)
 - **Method:** Implemented the Haversine formula using Python's math module
 - **Result:** Distance = **503.44 km**
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2. Third-party Modules

Objective: Repeat distance calculation using the geopy library.

- **Library Used:** geopy
- **Results:**

Method	Distance
Great Circle	503.45 km
Geodesic (WGS-84 ellipsoid)	501.10 km

3. Using Web APIs

Objective: Use OpenRouteService API to calculate driving distances.

Question 1: Hyderabad to Bengaluru

- **Driving Distance:** 569.05 km
- **Estimated Duration:** 27,527.6 seconds (~7.6 hours)

Question 2: San Francisco to Multiple Cities

Destination	Driving Distance
Los Angeles	615.56 km
Boston	4,991.47 km
Atlanta	3,976.33 km

4. Reading Files

Objective: Count the number of lines in the `worldcities.csv` file.

- **File:** `python_foundation/data/worldcities.csv`
 - **Result:** Total lines = **15,494**
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5. Reading CSV Files

Objective: Calculate geodesic distances from home city to all cities in the same country and export to CSV.

- **Home City:** Hyderabad, India
 - **Method:** Used `csv` module with `geopy.distance.geodesic()`
 - **Output File:** `Assignment2/cities_distance.csv`
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6. Working with Pandas

Objective: Use Pandas to filter and calculate distances, excluding the home city from results.

- **Home City:** Bengaluru, India
- **Method:** Used `pandas.DataFrame.apply()` with custom distance function
- **Result:** Calculated distances to **211 Indian cities**

Sample output:

City	Distance (km)
Mumbai	837.19
Delhi	1,738.64
Kolkata	1,552.64
Chennai	295.34
Hyderabad	500.05

Libraries Used

- `math` - Standard library for mathematical operations
 - `os` - File path operations
 - `csv` - CSV file handling
 - `geopy` - Geocoding and distance calculations
 - `requests` - HTTP requests for API calls
 - `pandas` - Data manipulation and analysis
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Key Learnings

1. **Haversine Formula** - Calculates great-circle distance assuming a spherical Earth
2. **Geodesic Distance** - More accurate calculation using Earth's ellipsoid shape (WGS-84)
3. **API Integration** - Using REST APIs to get real-world driving distances and durations
4. **Data Processing** - Efficient handling of large CSV datasets using both native Python and Pandas